

ESPERANCE AP, Australia

WMO: 956380

Lat: 33.6825S

Lon: 121.8275E

Elev: 468

StdP: 14.45

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.0	43.3	31.2	25.8	65.0	34.3	29.5	61.3	31.0	57.0	28.1	55.6	10.9	0	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.5	95.5	66.6	90.4	65.9	85.8	64.9	70.6	85.4	68.7	82.4	67.2	79.5	14.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	99.1	73.5	64.8	93.5	71.4	63.5	89.2	70.2	34.8	85.6	33.2	82.5	31.9	79.4	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.0	24.0	21.8	DB	38.7	106.4	1.1	4.5	38.0	109.6	37.3	112.2	36.7	114.7	36.0	117.9	(4)
(5)			WB	37.3	74.7	1.3	2.4	36.4	76.4	35.6	77.8	34.9	79.2	34.0	80.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.3	69.2	69.7	67.3	63.9	59.1	54.9	53.2	54.1	56.1	59.3	63.3	66.3	(6)	
	DBStd	7.77	6.13	5.72	5.88	5.45	4.58	3.36	3.28	4.19	4.75	5.80	6.27	6.08	(7)	
	HDD50	18	0	0	0	0	0	2	7	6	3	0	0	0	(8)	
	HDD65	1980	17	12	37	85	193	302	365	339	272	198	109	51	(9)	
	CDD50	4153	596	551	538	418	282	150	107	133	186	289	398	505	(10)	
	CDD65	641	148	144	110	53	11	0	0	1	4	22	57	91	(11)	
	CDH74	5858	1396	1185	892	373	73	2	0	17	55	270	637	958	(12)	
	CDH80	2517	677	520	376	109	13	0	0	2	13	91	266	450	(13)	
(14) Wind	WSAvg	11.9	12.6	12.3	11.4	10.4	11.0	11.8	12.3	12.0	12.1	11.7	12.2	12.5	(14)	
(15) Precipitation	PrecAvg	21.6	0.9	1.1	1.3	1.5	2.7	2.7	3.2	2.8	2.2	1.8	1.5	0.7	(15)	
	PrecMax	28.7	8.2	3.8	6.9	3.6	5.7	7.4	5.6	5.0	5.5	3.3	4.2	2.3	(16)	
	PrecMin	12.8	0.0	0.0	0.0	0.2	0.5	1.1	0.7	0.8	0.0	0.5	0.1	0.0	(17)	
	PrecStd	3.5	1.5	1.0	1.4	0.9	1.3	1.3	1.1	1.1	1.2	0.8	0.9	0.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	103.2	99.9	96.6	89.3	81.3	71.2	69.5	76.0	80.8	89.6	95.7	100.0	(19)	
		MCWB	68.7	68.6	65.7	63.9	60.6	55.5	55.2	59.0	60.9	61.8	64.4	65.1	(20)	
	2%	DB	95.2	91.9	89.6	82.7	74.9	67.5	65.6	69.3	73.5	81.2	87.8	91.1	(21)	
		MCWB	67.4	67.8	65.4	62.7	58.8	55.4	54.7	56.5	58.3	61.0	63.7	65.1	(22)	
	5%	DB	87.5	86.4	83.3	77.9	71.0	64.7	62.8	65.4	69.2	75.4	81.4	84.5	(23)	
		MCWB	66.7	66.7	64.3	61.7	58.3	54.7	53.9	55.4	57.4	59.8	62.1	64.6	(24)	
	10%	DB	81.4	81.0	77.9	73.7	67.5	62.2	60.3	62.2	65.3	70.4	75.6	78.3	(25)	
		MCWB	65.5	66.0	63.9	61.0	57.2	53.7	53.1	54.1	55.8	58.4	61.0	63.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.2	74.1	70.6	68.6	65.4	60.6	59.4	61.2	62.9	65.7	68.5	70.3	(27)	
		MCDB	89.9	87.2	83.5	79.0	74.1	66.6	63.5	70.4	74.8	79.4	86.0	84.7	(28)	
	2%	WB	70.3	70.9	68.1	65.6	61.9	58.2	56.8	58.6	60.3	62.8	65.7	67.5	(29)	
		MCDB	87.0	85.1	80.4	75.4	68.7	63.2	62.1	65.6	69.2	75.9	80.1	83.8	(30)	
	5%	WB	68.2	68.7	66.5	63.9	60.2	56.5	55.2	56.6	58.3	60.9	63.7	65.6	(31)	
		MCDB	82.6	80.8	76.7	72.9	66.9	61.6	60.4	63.1	66.2	72.0	77.0	80.4	(32)	
	10%	WB	66.4	67.0	64.9	62.5	58.6	55.1	53.8	55.0	56.7	59.2	61.9	63.8	(33)	
		MCDB	78.2	77.4	73.9	70.1	64.8	60.2	58.9	60.9	63.9	68.5	73.3	75.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	20.4	19.5	18.7	18.1	15.8	14.4	14.1	15.7	17.7	19.1	20.0	19.8	(35)	
		MCDBR	33.5	30.6	28.8	24.9	21.4	18.4	18.4	21.3	24.8	29.6	31.4	32.7	(36)	
		MCWBR	11.3	10.3	9.3	9.7	9.7	10.0	10.4	11.8	12.1	12.7	11.9	12.1	(37)	
	5% WB	MCDBR	26.5	24.7	21.7	19.7	16.8	15.3	15.1	18.9	20.9	24.3	25.8	27.0	(38)	
		MCWBR	10.7	10.2	9.1	9.6	9.5	10.0	9.9	11.5	11.5	11.9	11.3	11.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.354	0.348	0.329	0.320	0.301	0.292	0.293	0.302	0.315	0.332	0.337	0.336	(40)	
	taud		2.520	2.551	2.592	2.609	2.633	2.628	2.622	2.585	2.542	2.511	2.522	2.548	(41)	
	Ebn at Noon		312	307	302	288	278	272	278	290	302	309	315	319	(42)	
	Edh at Noon		36	33	30	26	23	21	22	26	31	34	35	35	(43)	
(44) All-Sky Solar Radiation	RadAvg		2332	2035	1675	1260	932	798	865	1149	1540	1910	2194	2406	(44)	
	RadStd		126	139	109	63	55	30	46	55	61	120	143	147	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-108	+167	+79

Nomenclature: See separate page